Title: Data Cleaning and Preprocessing using Python

Objective

The objective of this task is to clean and preprocess raw data to make it suitable for analysis by removing inconsistencies, handling missing values, and improving data quality.

Dataset Description

The dataset used contains structured data in CSV format with multiple columns representing different attributes. The dataset may contain missing values, duplicate entries, and inconsistent formats.

Tools Used

- Python
- Pandas Library
- Google Colab

Steps Performed

1. Data Loading

The dataset was loaded into Python using the pandas library.

```
df = pd.read_csv("hw_200.csv")
```

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2. Data Inspection

- Checked first few rows using `head()`
- Checked structure using `info()`
- Checked missing values using `isna().sum()`

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3. Removing Duplicates

Duplicate rows were identified and removed.

```
df = df.drop_duplicates()
```

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4. Handling Missing Values

- Numerical columns → filled using median
- Categorical columns → filled using mode

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5. Data Standardization

- Converted text columns to lowercase
- Removed extra spaces

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6. Saving Cleaned Data

The cleaned dataset was saved as:

```
df.to_csv("cleaned_data.csv", index=False)
```

Outcome

The dataset was successfully cleaned and transformed into a structured format suitable for analysis. The cleaned dataset was saved as **cleaned_data.csv**.

Conclusion

Data cleaning is an essential step in data science as it improves data quality and ensures accurate analysis. This task helped in understanding real-world data issues and applying appropriate preprocessing techniques.